25901262

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAREXAMINATION March2026
Fourth SEMESTER
Discipline Specific Core Course (DSC) - MG4DSCFSQ201 - INTRODUCTION TO FOOD MICROBIOLOGY
(2024 ADMISSION ONWARDS)
Duration: 1 hour 30 minutes Maximum Marks: 50

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
MCQ
Answer all questions
Each question carries **1**marks

1. Identify the part of eukaryotic cell called the "powerhouse" of the cell.
Answer: c) Mitochondria

2. Select the correct covering found in protozoa like *Paramecium*.
Answer: c) Pellicle

3. Define pure culture.
Answer: c) Population of cells arising from a single cell

4. Define physical sterilization.
Answer: c) Sterilization using physical agents

5. *Vibrio cholerae* infection is mainly transmitted through
Answer: b) Contaminated water and food

6. Soft swell has
Answer: b) Distorted ends of the can

7. Identify the food which is most susceptible to aflatoxin contamination.
Answer: b) Nuts and oilseeds

8. Soft rot of vegetables is mainly caused by
Answer: b) Bacterial pectinolytic enzymes

9. Deterioration of fatty fish produces appreciable amounts of 'stale fishy' smell, which is due to _____.
Answer: d) Trimethylamine oxide

10. Putrefaction is mainly associated with degradation of
Answer: c) Proteins

[1x10 = 10]

Part B
Short Answer
Answer any 4 questions
Each question carries 5 marks

11. State the contributions of Robert Koch and outline Koch's postulates.
Answer
Contributions of Robert Koch

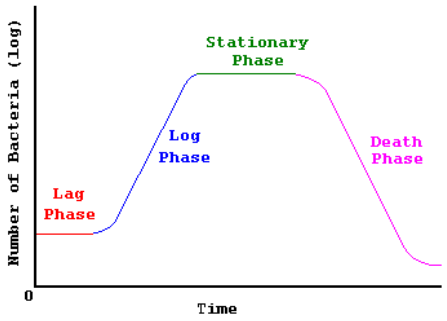
Robert Koch is known as the **Father of Medical Microbiology**. His major contributions include:

- Established the germ theory of disease.
- Identified *Bacillus anthracis* as the cause of anthrax.
- Discovered *Mycobacterium tuberculosis*, the causative agent of tuberculosis.
- Identified *Vibrio cholerae* as the causative organism of cholera.
- Developed pure culture techniques.
- Introduced staining methods and solid culture media.

Koch's Postulates

1. The microorganism must be present in all diseased individuals and absent from healthy individuals.
2. The microorganism must be isolated and grown in pure culture.
3. The pure culture should produce the same disease when inoculated into a healthy host.
4. The microorganism must be re-isolated from the experimentally infected host and shown to be identical to the original organism.

12. Describe the different phases of bacterial growth curve.



The bacterial growth curve consists of four phases:

1. Lag Phase

- Cells adapt to the new environment.
- No increase in cell number.
- Active synthesis of enzymes and proteins.

2. Log (Exponential) Phase

- Rapid multiplication occurs.
- Cells divide at a constant rate.
- Maximum metabolic activity is observed.

3. Stationary Phase

- Nutrients become limited.
- Waste products accumulate.
- Rate of cell growth equals rate of cell death.

4. Death Phase

- Cells die faster than they are produced.
- Population declines exponentially.
- Caused by nutrient depletion and toxic metabolites.

13. Identify the structural components of a typical virus and state the function of each.

- **Draw the Structucture and explain**

Structural Component	**Function**
Nucleic acid (DNA/RNA)	Carries genetic information
Capsid	Protects nucleic acid
Capsomeres	Protein subunits forming capsid
Envelope	Provides protection and aids infection
Surface spikes	Attachment to host cells
Matrix proteins	Stabilize virus structure

Functions

- Genetic material controls replication.
- Capsid protects viral genome.
- Envelope helps entry into host cells.

- Spikes recognize host cell receptors.

14. Describe the principle and procedure of spread plate method.

Principle

The spread plate method is used to isolate and enumerate microorganisms. A known volume of diluted sample is spread over the surface of solid agar medium. Each viable cell develops into a separate colony.

Procedure

1. Prepare serial dilutions of the sample.
2. Transfer 0.1 mL of diluted sample onto sterile agar plate.
3. Spread evenly using a sterile L-shaped glass spreader.
4. Allow inoculum to absorb.
5. Incubate at suitable temperature.
6. Count colonies formed.

Applications

- Enumeration of microorganisms.
- Isolation of pure cultures.
- Food and water microbiological analysis.

15. Define putrefaction and explain its significance in meat spoilage.

Putrefaction is the microbial decomposition of proteins present in meat, resulting in the formation of foul-smelling compounds.

Significance in Meat Spoilage

- Produces ammonia, hydrogen sulfide, indole and skatole.
- Causes offensive odours.
- Alters flavour and texture.
- Reduces nutritional quality.
- Makes meat unsuitable for consumption.

16. Describe the role of moisture in spoilage of stored cereal grains.

Moisture plays a major role in grain spoilage.

Effects of High Moisture

- Promotes growth of moulds and bacteria.
- Increases respiration of grains.
- Causes heating and caking.
- Encourages insect infestation.
- Leads to production of mycotoxins such as aflatoxins.
- Reduces shelf life and quality.

Prevention

- Maintain grain moisture below 13–14%.
- Store in dry and well-ventilated conditions.

[5x4 = 20]

Part C
Essay
Answer any 2 questions
Each question carries 10 marks

17. Recall the methods of sexual reproduction in fungi.

Sexual reproduction in fungi involves fusion of compatible nuclei and production of genetically diverse offspring.

Stages of Sexual Reproduction

1. Plasmogamy
Fusion of cytoplasm of two compatible cells.
2. Karyogamy
Fusion of nuclei.
3. Meiosis
Reduction division producing haploid spores.
Methods of Sexual Reproduction
A. Planogametic Copulation
Fusion of motile gametes.
Example: Lower fungi
B. Gametangial Contact
Male and female gametangia come into contact and nuclei migrate through a fertilization tube.
Example:*Pythium*
C. Gametangial Copulation
Entire gametangia fuse and their contents unite.
Example:*Rhizopus*
D. Spermatization
Non-motile male cells (spermatia) fuse with receptive hyphae.
Example: Rust fungi
E. Somatogamy
Fusion of ordinary vegetative cells.
Example: Mushrooms
Importance

- Generates genetic variation.
- Helps adaptation.
- Produces resistant spores.

18. Explain any five chemical sterilizing agents and summarize the factors that influence their effectiveness.

Chemical sterilizing agents destroy microorganisms by damaging proteins, membranes and nucleic acids.
1. Ethylene Oxide

- Gas used for heat-sensitive materials.
- Alkylates proteins and DNA.

2. Formaldehyde

- Used for room fumigation.
- Denatures proteins.

3. Glutaraldehyde

- High-level disinfectant.
- Used for endoscopes and surgical instruments.

4. Hydrogen Peroxide

- Strong oxidizing agent.
- Destroys cellular components.

5. Alcohols (70% Ethanol)

- Denature proteins.
- Dissolve membrane lipids.

Factors Affecting Effectiveness

- Concentration of sterilant.
- Contact time.
- Temperature.
- pH.
- Presence of organic matter.
- Number of microorganisms.
- Type of microorganism.
- Humidity and moisture conditions.

19. Explain *Vibrio cholerae* and Cholera in detail.

Vibrio cholerae is a Gram-negative bacterium responsible for cholera, a severe diarrheal disease.

Characteristics

- Gram-negative curved rod.
- Motile with polar flagellum.
- Non-spore-forming.
- Facultative anaerobe.

Sources

- Contaminated water.
- Contaminated food.
- Poor sanitation systems.

Transmission

Occurs through the fecal-oral route by consuming contaminated food and water.

Pathogenesis

1. Bacteria enter intestine.
2. Colonize intestinal mucosa.
3. Produce cholera toxin.
4. Stimulate secretion of water and electrolytes.
5. Cause severe watery diarrhea.

Symptoms

- Rice-water stools.
- Vomiting.
- Muscle cramps.
- Severe dehydration.
- Electrolyte imbalance.
- Shock in severe cases.

Diagnosis

- Stool culture.
- Microscopic examination.
- Serological tests.

Prevention

- Safe drinking water.
- Proper sanitation.
- Personal hygiene.
- Food safety measures.
- Vaccination.

Treatment

- Oral Rehydration Solution (ORS).
- Intravenous fluids.
- Antibiotics in severe cases.

Public Health Importance

Cholera remains a major water-borne disease worldwide and is associated with poor sanitation and contaminated water supplies.

20. Describe the spoilage of milk with reference to acid, proteolytic, and gas-forming spoilage.

Answer

Introduction

Milk is a highly nutritious food and serves as an excellent medium for microbial growth, leading to spoilage.

A. Acid Spoilage

Causative Organisms

- *Lactobacillus*
- *Lactococcus*
- *Streptococcus*

Process

Lactose is converted into lactic acid.

Effects

- Sour taste.
- Increased acidity.
- Coagulation of casein.
- Curd formation.

B. Proteolytic Spoilage

Causative Organisms

- *Pseudomonas*
- *Bacillus*
- *Clostridium*

Process

Proteins are broken down by microbial enzymes.

Effects

- Bitter taste.
- Putrid odour.
- Texture deterioration.

C. Gas-forming Spoilage

Causative Organisms

- Coliform bacteria
- *Clostridium* species
- Certain yeasts

Process

Fermentation of lactose produces carbon dioxide and hydrogen.

Effects

- Swollen containers.
- Foaming.
- Unpleasant taste and odour.

Prevention of Milk Spoilage

- Proper pasteurization.
- Refrigerated storage.
- Hygienic handling and packaging.

[10x2 = 20]